INCH-POUND

MIL-C-24643/37C
14 March 1994
SUPERSEDING
MIL-C-24643/37B
3 October 1986

MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, TYPE LS3U
(INCLUDING VARIATION LS3UA)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-24643.

This cable can be manufactured in two variations: LS3U (unarmored) and LS3UA (armored).

REQUIREMENTS:

Qualification required.

Construction (non-watertight)

First - Copper conductor, AWG no. 18, class B stranding, (7 by 0.0152 inch) uncoated. Maximum diameter 0.050 inch.

Second - Separator may be used at manufacturer's option where required to provide free-stripping insulation.

* Third - Crosslinked polyethylene insulation. Nominal wall thickness 0.023 inch; wall thickness may vary from nominal as necessary, in order for completed cable to meet the specified electrical requirements. Colored insulation, one black, one white and one red conductor for each triad.

Fourth - Three conductors (one black, one white and one red) cabled together with a nominal lay of 3.5 inches to form each triad.

* Fifth - Identification braid on each triad. Standard identification code applied by method 4.

Sixth - The specified number of triads (see table I) cabled together with a lay not greater than 24 times the pitch diameter of the layer. Cabling sequence shall be consecutive, starting with no. 1, from the center outward. Fillers shall be used as necessary to make a firm, well rounded assembly.

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Construction (non-watertight) (continued)

Seventh - Binder tape of polyester, type G in accordance with MIL-I-631, applied helically with overlap.

Constructive variant for type LS3U

* Eighth - Cross-linked polyolefin jacket. (see table I for thickness).

Constructive variant for type LS3UA

* Eighth - Cross-linked polyolefin jacket. (see table I for thickness).

* Ninth - Braided metal armor.

* TABLE I. Details.

Military 1/ part no. M24643/37	Size 2/	Number of triads	Cable jacket thickness min avg (inch)	Overall diameter (inches)			
				TYPE LS3U		TYPE LS3UA	
				min	max	min	max
-01UN	3	3	0.040	0.580	0.620	0.630	0.670
-02UN	7	7	.040	.760	.810	.810	.860
-03UN	12	12	.050	1.030	1.090	1.080	1.140

1/ Unarmored. When armor option is required, see MIL-C-24643 for configuration of part number.
2/ Type LS3U and LS3UA.

EXAMINATION AND TESTS:

	Requirements
Basic electrical:	
Conductor Resistance - ohms/1000 feet at 25°C, maximum	7.06
Voltage withstand - volts, root mean square, minimum	
Conductor to conductor	2000
Insulator resistance - Megohms/1000 feet minimum	
Conductor to conductor	500
Conductor continuity	No failure
Group A:	
Visual and dimensional	No failure
Capacitance	
Mutual capacitance - 1 kilohertz (KHz), pF/ft, maximum	30
Capacitance unbalance - percent maximum	8

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EXAMINATION AND TESTS (continued):

	Requirements
Group B:	
* Cross-linked proof test (percent, maximum)	
Insulation	50
Jacket (When tested at 200°C)	50
Tear - pounds per inch thickness, minimum (ASTM D 470)	35
Physicals (unaged)	
Insulation	
Tensile strength - lb/in^2, minimum	1800
Elongation - percent, minimum	250
Jacket (cable)	
Tensile strength - lb/in^2, minimum	1300
Elongation - percent, minimum	160
Group C:	
Physicals (aged)	
Insulation	
Air oven	
Tensile strength - percent of unaged, minimum	80
Elongation - percent of unaged minimum	80
Jacket (cable)	
Air oven	
Tensile strength - percent of unaged, minimum	60
Elongation - percent of unaged minimum	60
Hot oil immersion	
* Tensile strength - percent of unaged, minimum	50
* Elongation - percent of unaged, minimum	50
Shrinkage	No failure
Heat distortion - percent of unaged, maximum	30
* Permanence of printing (jacket) - (type LS3U only) cycles, minimum	125
Armor (type LS3UA only) - conformance to material, construction and coverage	No failure
* Group D:	
Flame propagation (cable)	No failure

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QUALIFICATION INSPECTION:

* Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:

	Requirements
Aging and compatibility (cable) (125± 5°C)	No failure
Abrasion resistance (jacket) - scrapes, minimum	75
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Immersion (jacket)	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
Smoke index, maximum	
Jacket	25
* Fillers	45
Insulation	45
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - (type A) hardness, minimum	80
Weathering (jacket)	No failure

UNIT ORDERING LENGTHS: All sizes 1000 feet nominal

NOTE: The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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(Project No. 6145-2057-37)

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